

Prompt setups. We test three prompt settings for MMLU: (1) “zero-shot”: a zero-shot prompt; (2) “TTTTT”: a five-shot prompt with all correct exemplars; (3) “TTFFF”: a five-shot prompt where the first two examples are correct and the following three are incorrect. For TriviaQA, we test 5-shot and 20-shot prompts.

Although it is true that the correctness of few-shot examples could be verified in practice, our motivation to study incorrect in-context examples includes: (1) It helps understand how truthfulness probes handle context that contains mixed truthful and untruthful information; (2) It provides insights into how robust truthfulness probes are to potentially conflicting information.

4.5.2 Results

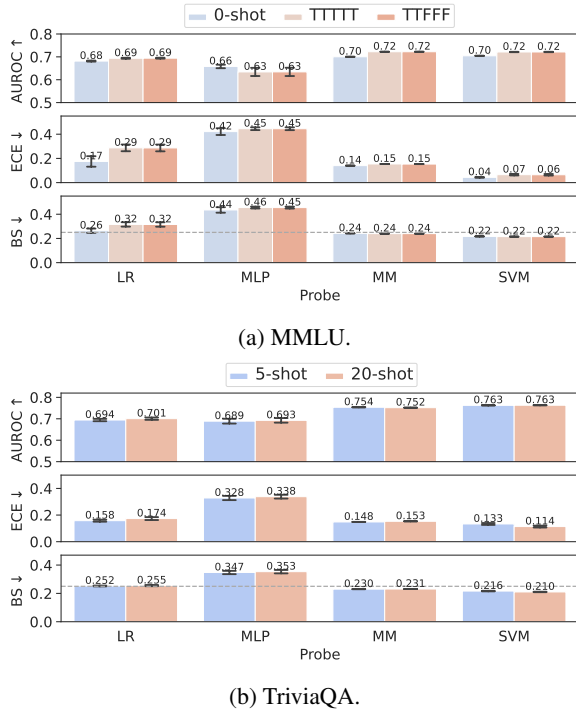


Figure 5: AUROC↑/ECE↓/BS↓ of truthfulness probes for Llama-3.1-8B on MMLU and TriviaQA. The dashed gray line corresponds to random results, and error bars denote standard error.

The results, shown in Figure 5, reveal that across all probes, accuracy generally improves when few-shot exemplars are provided. This suggests that providing more task-related context in the prompt typically aids the LLM in the implicit truthfulness classification task, and that truthfulness probes not only generalize from factual statements to both multiple-choice QA and short-form QA, but also generalize from fundamental commonsense knowl-

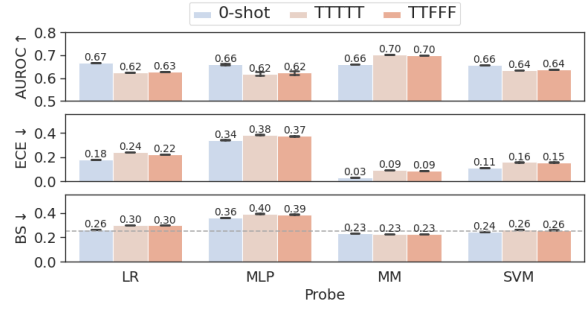


Figure 6: AUROC↑/ECE↓/BS↓ of in-domain probes for Llama-3.1-8B trained and tested on MMLU.

edge to both expert knowledge (MMLU) and trivia knowledge (TriviaQA). Notably, according to the results on MMLU, the effect of imperfect few-shot prompts is near identical to entirely correct few-shot prompts. This indicates that the performance of truthfulness probes is not significantly influenced by the truthfulness of the few-shot examples – the probes express the truthfulness of the *final* (Q, A) pair, treating prior exemplars as context. Interestingly, this finding aligns with Halawi et al. (2023)’s observation that early layers of the LLM are insensitive to false in-context demonstrations.

However, few-shot prompting does not always improve calibration, and a positive case is only observed for the SVM probe on the TriviaQA dataset. Among all probes, the SVM probe performs best in terms of both classification accuracy and calibration, likely because of its ability to accurately identify the geometry of the truth direction and the benefits of the additional Platt scaling procedure.

4.5.3 Additional Experiments

Beyond the generalization of probes trained on factual statements, we are interested in how these probes differ from in-domain probes, which are trained and tested on data of the same domain. Therefore we perform follow-up experiments by training and testing probes on MMLU using the Llama-3.1-8B model, using the same data and prompt setup as above. Specifically, we train on a random 70% split and test on the rest 30%.

We present results averaged over three random trials in Figure 6. Comparing the results with those of Figure 5, we observe that in-domain probes generally under-performs truthfulness probes trained on atomic factual statements in terms of AUROC. This finding indicates that truthfulness probes under the current setup perform better than those under the in-domain QA setup.

4.6 Contextual Knowledge

When training truthfulness probes using factual statements, we are targeting the factual aspect of truthfulness, where the grounding knowledge resides in the LLM’s parameters. However, in-context knowledge also plays a vital role in generation. In this section, we investigate if truthfulness probes can also capture this additional aspect of truthfulness, where the grounding knowledge is provided as contextual information in the prompt. Notably, context grounding is fundamentally different from factual correctness, since faithfully following false context is acceptable for the former but not for the latter. We conduct experiments on two tasks: in-context QA and abstractive summarization.

4.6.1 Experimental Setup

Data. The training data consists of all the atomic statements. For the in-context QA task we use the SciQ (Welbl et al., 2017) and BoolQ (Clark et al., 2019) datasets. For SciQ, we randomly select 1000 questions, pairing each question with the lettered choice for both the true and a randomly selected false answer. The answers of BoolQ are binary “yes/no”, therefore we flip the answers to balance true and false (Q, A) pairs. For the abstractive summarization task, we use the XSum dataset (Narayan et al., 2018) and XSum Hallucination Annotations (Maynez et al., 2020). For each article, we pair true summaries from the former dataset with false ones from the latter one.

Models. We use Llama-3.1-8B.

Methods. We apply the four probe instantiations introduced in Section 3.2.

Metrics. We report AUROC, ECE and BS. The results are averaged over three trials, with randomness introduced via initialization and data splits.

Prompt setups. For SciQ dataset which is a multiple-choice task, we implement four settings: (1) “zero-shot”: zero-shot prompt; (2) “TTT”: three-shot prompt where all exemplars are correct; (3) “TTF”: three-shot prompt, where the first two examples are correct and the third is incorrect; (4) “FFT”: three-shot prompt, where the first two examples are incorrect and the third is incorrect. For BoolQ, we implement four settings: “no options”, “with options”, and one-shot “T”/“F” (with possible options). For XSum, no options are provided in the prompt as it is not a multiple-choice task. We

implement the following prompt configurations: zero-shot, “T”, “TT”, and “TTT”.

4.6.2 Results

The results are shown in Figure 7. Across all datasets and most probes, possible answer options and few-shot exemplars generally improve classification accuracy. This indicates that truthfulness probes generalize from factual statements to both in-context multiple-choice QA tasks and abstractive summarization tasks.

Among the probes, statistics-based probes (LR and MLP) display greater standard error in terms of all three metrics than geometric-oriented ones (MM and SVM), likely due to their optimization instability, and the SVM probe performs best from the perspective of BS. Additionally, discrepancies are observed for LR probe on the SciQ and BoolQ datasets, MLP probe on the BoolQ dataset, and MM probe on the XSum dataset, where accuracy improves in response to in-context exemplars but calibration worsens. We assume that these discrepancies could be explained with the overconfidence of the probes’ predictions.

4.7 Selective Question Answering

Based on the findings of Section 4.5, we observe that truthfulness probes trained on atomic statements are capable of generating calibrated probabilistic predictions for QA tasks while achieving reasonable accuracy. Building upon these observations, this section investigates whether truthfulness probes can be leveraged to selectively identify correct answers from a set of candidate responses sampled from an LLM. This setup is inspired by the work of Kadavath et al. (2022), who demonstrated that an LLM can evaluate the correctness of its own answers through verbal feedback. While our selective QA experiment shares similarities with Kadavath et al. (2022), our approach differs fundamentally as we leverage internal representations rather than model-generated feedback.

For this experiment, we use the TriviaQA test data from Section 4.5, where we sample 20 answers from the Llama-3.1-8B model using a 20-shot prompt with unit temperature. To perform selective QA, we select the subset of (Q, A) pairs for which the truthfulness probe predicts $P[\Phi = 1] > 0.5$ and report the accuracy on this subset. We use the SVM probe, as it performs best in terms of both classification accuracy and calibration.

The aggregated accuracy across all sampled

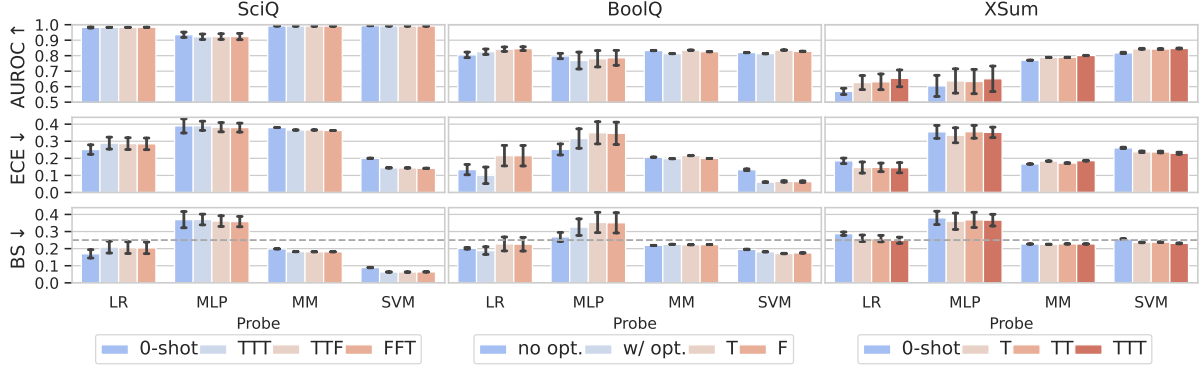


Figure 7: AUROC↑/ECE↓/BS↓ of truthfulness probes for Llama-3.1-8B on tasks where grounding knowledge is provided in the prompt. The dashed gray line corresponds to random results, and error bars denote standard error.

(Q, A) pairs is 55.29%. Among these, the truthfulness probe classifies 80.26% as true, and the accuracy of this subset is 64.06%. This demonstrates that truthfulness probes can be used to filter out false answers sampled from LLMs.

5 Conclusions

In this work, we provide preliminary evidence supporting our hypothesis that consistent truth directions only emerge in capable LLMs and not in weaker ones, and they could be effectively identified with simple linear probes. We also investigate the generalization properties of truth directions. Empirical results show that truthfulness probes trained only on atomic statements generalize well to logical transformations, (few-shot) question answering and contextual truthfulness. These findings underscore the potential of truthfulness probes to identify truth directions using simple anchor data, thereby facilitating the elicitation of latent knowledge within LLMs.

6 Limitations

This study has several limitations that warrant discussion. First, the term "truth" as used in this paper represents an idealized concept, but it may not be what the truthfulness probes actually measure. Drawing on Kadavath et al. (2022) and Marks and Tegmark (2023), the pretraining of language models largely involves imitating human-generated text. Consequently, truthfulness probes are likely capturing an overlap between widely accepted human beliefs and factual, objective truths about the physical world. This raises interesting questions about how such probes might perform in the context of scalable oversight, particularly with hypothetical AI systems that surpass human intelligence.

Second, our investigation into the generalization of truthfulness probes is limited to short-form QA. Extending this analysis to more complex scenarios, such as long-form QA or instruction-following tasks, may yield novel insights and uncover more practical applications of truthfulness probes.

Third, the causality of truthfulness probes is unclear. Our approach relies on classic classification techniques, consistent with prior work on probing truthfulness. Meanwhile, we do not discuss the causal effects of truth directions, i.e., whether LLMs utilize the implicit truthfulness classification results for predictions. Marks and Tegmark (2023) conducted causal intervention experiments using the mass-mean probe and showed that mass-mean directions are highly causal. However, as highlighted by Kumar et al. (2022), probes often capture spuriously-correlated features rather than exclusively isolating the target feature. Future research on the causal implications of truth directions could provide valuable insights into guiding LLMs to produce more truthful responses, as in Li et al. (2023).

Finally, our experiments are constrained by computational resources, with the largest model evaluated being Llama-3.1-70B. As a result, our hypothesis that highly capable LLMs will eventually establish a consistent internal concept of truthfulness remains untested on more advanced models such as GPT-4 (Achiam et al., 2023).

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